DPO and GRPO

RL without a critic, and preferences without RL

Carlos Cotrini

ETH Zurich

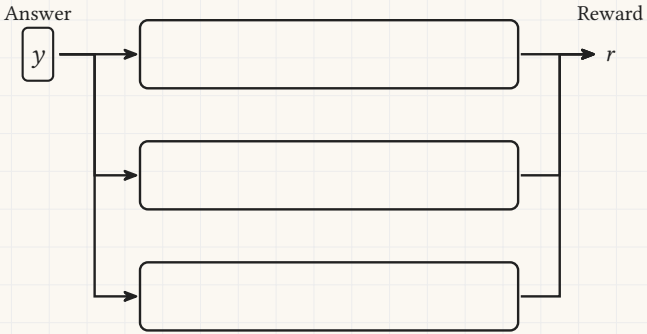
20 October 2026

1

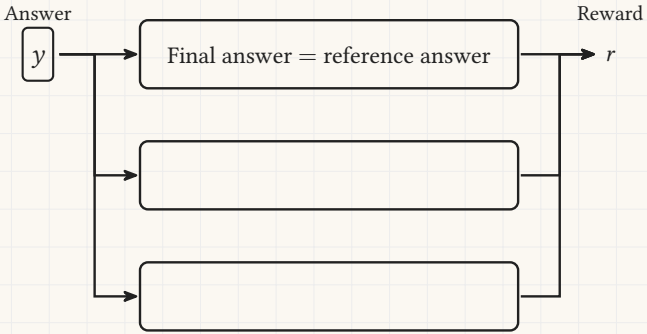
Section 1

Verifiable rewards

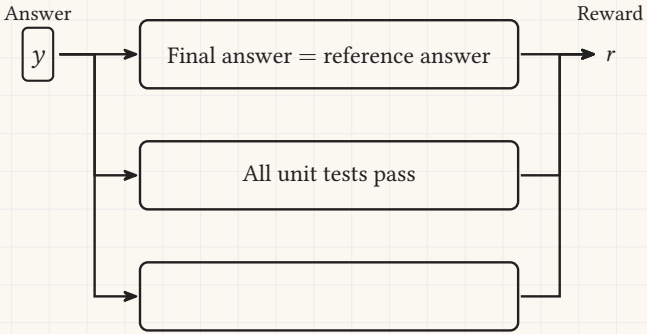
Rewards a program can check



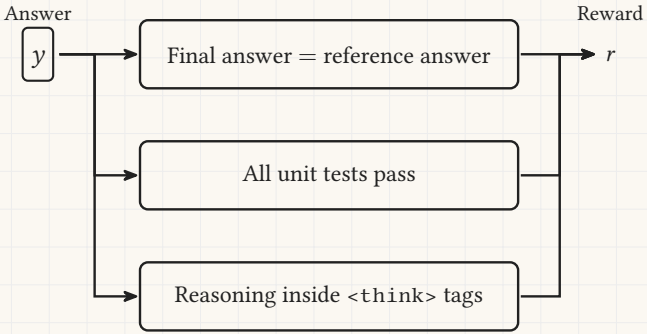
Rewards a program can check



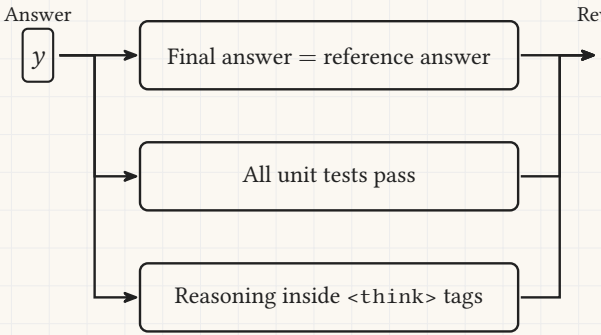
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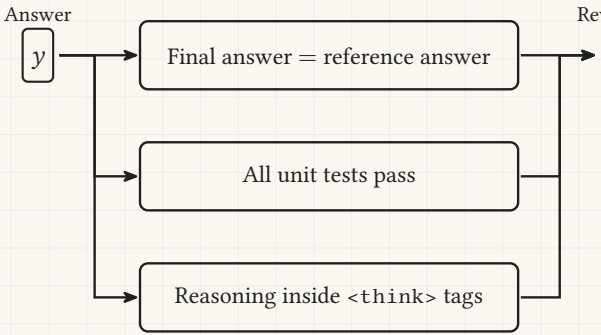
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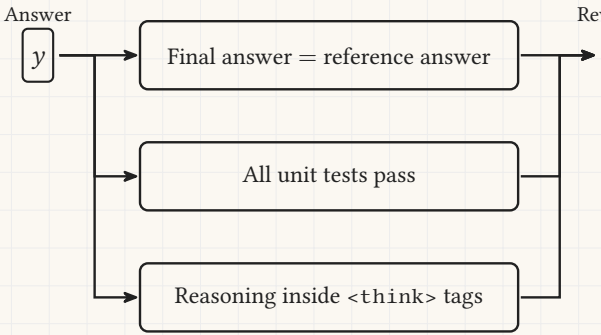


Rewards a program can check



$r \in \{0, 1\}$	Liu et al. 2025
$\in \{-1, 1\}$	Yu et al. 2025
$\in \{0, 10\}$	Lambert et al. 2025
$=$ accuracy $+$ format	Guo et al. 2025

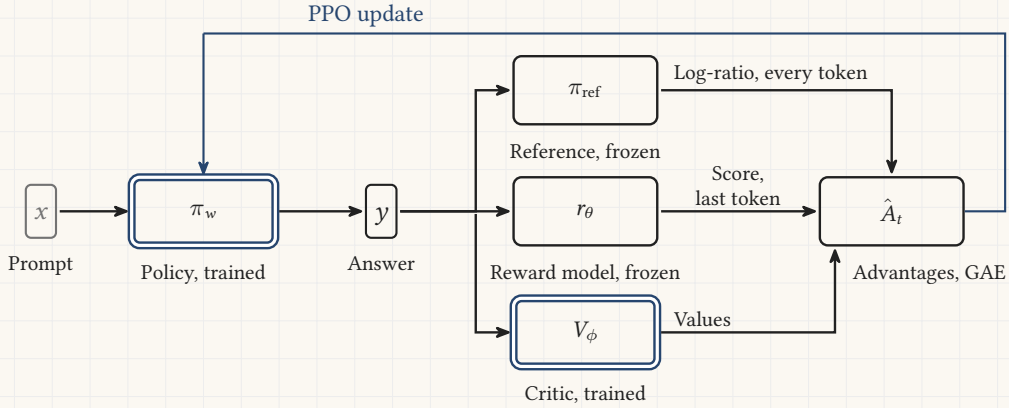
Rewards a program can check



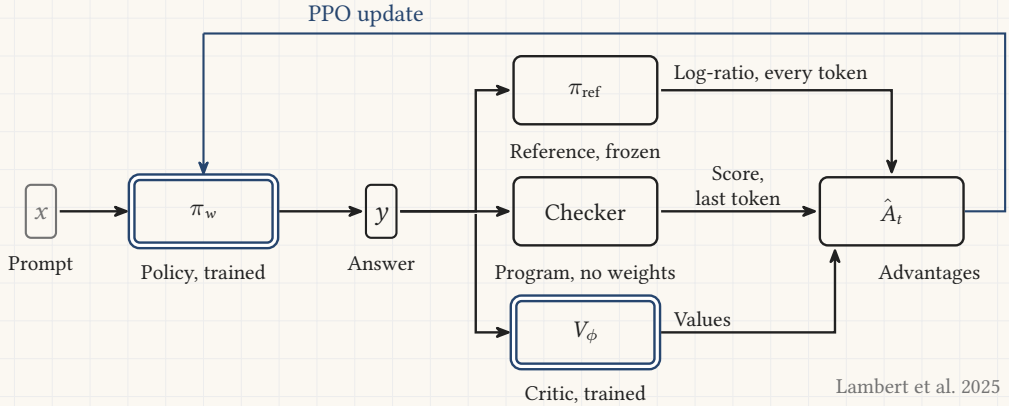
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DeepSeek-R1: learned reward models get hacked; retraining costs compute.

The same loop with a checker

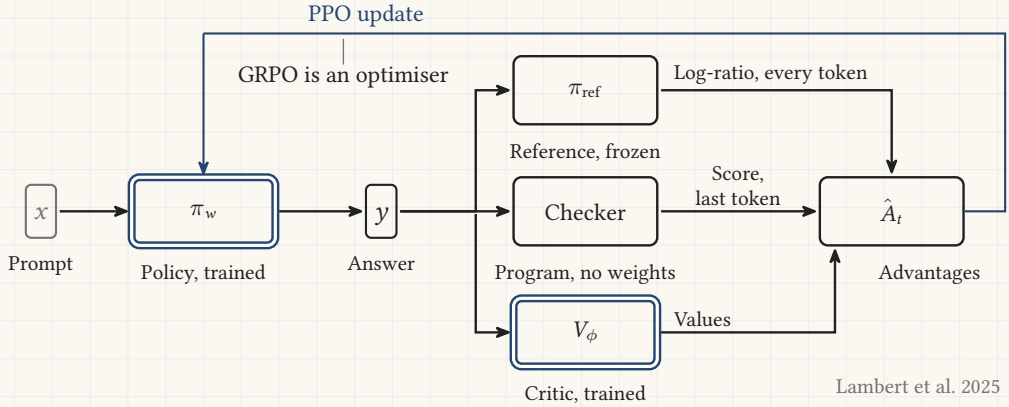


The same loop with a checker



Lambert et al. 2025

The same loop with a checker



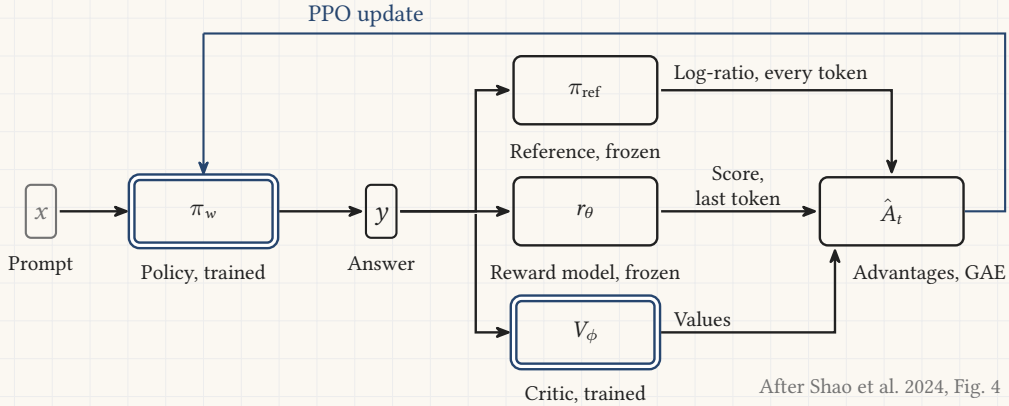
Lambert et al. 2025

2

Section 2

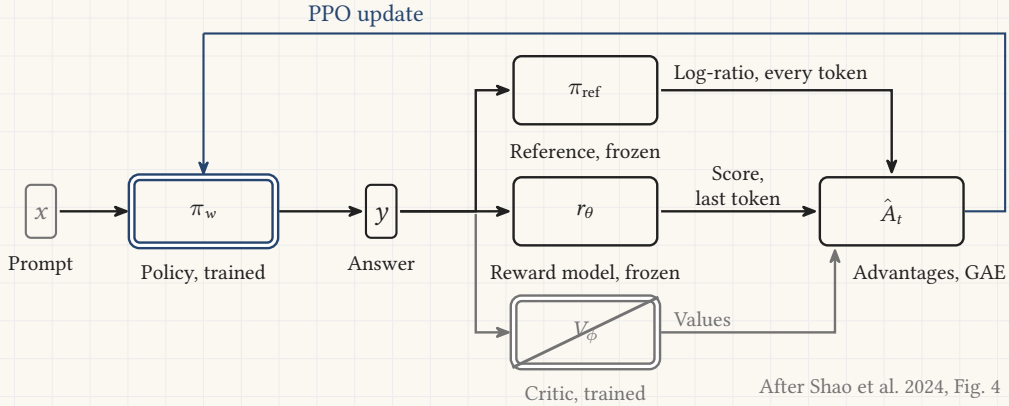
GRPO

From PPO to GRPO



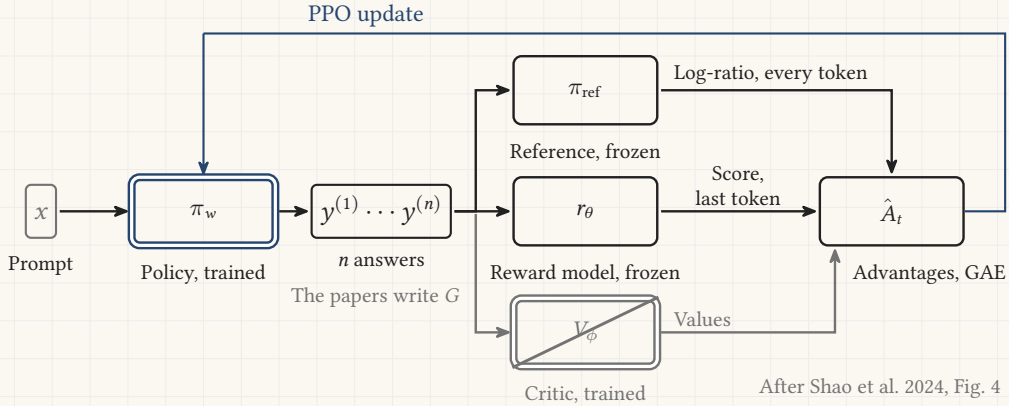
After Shao et al. 2024, Fig. 4

From PPO to GRPO



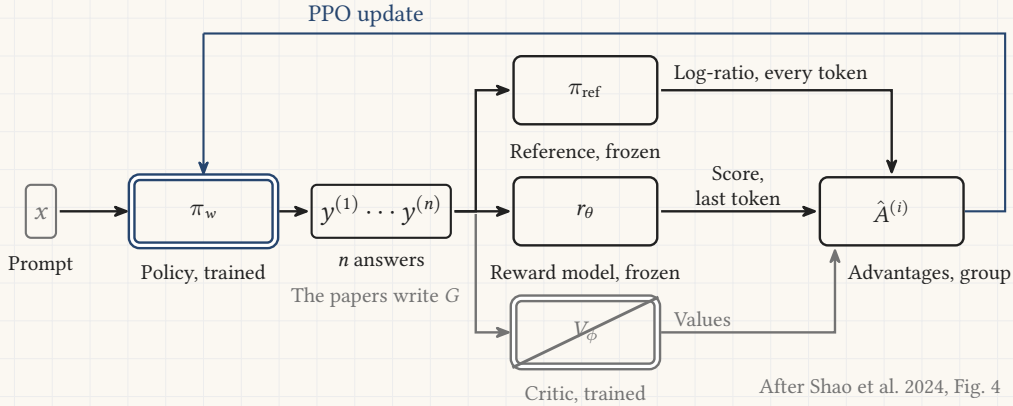
After Shao et al. 2024, Fig. 4

From PPO to GRPO

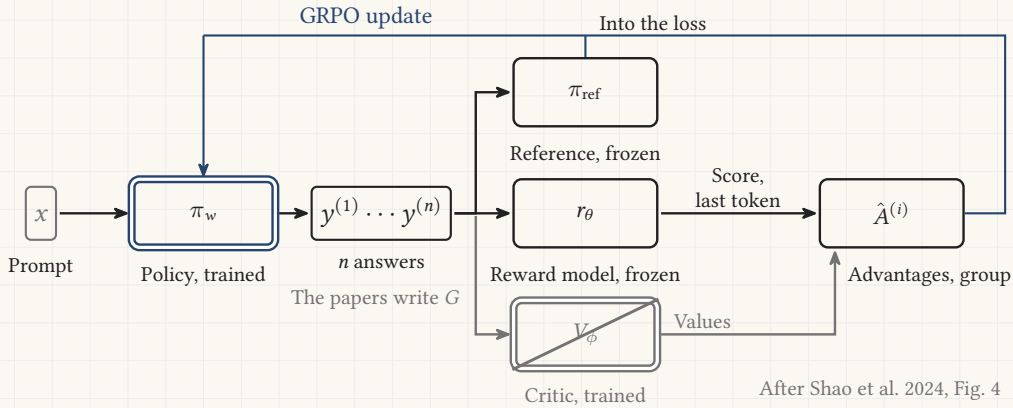


After Shao et al. 2024, Fig. 4

From PPO to GRPO

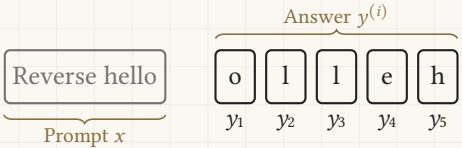


From PPO to GRPO

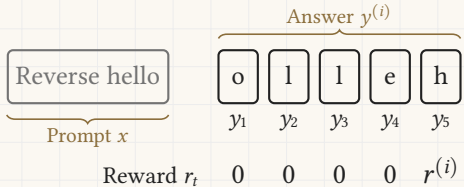


After Shao et al. 2024, Fig. 4

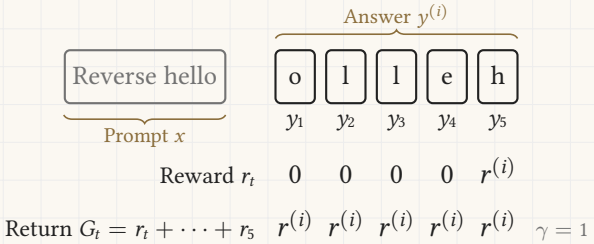
Every token's return is the answer's reward



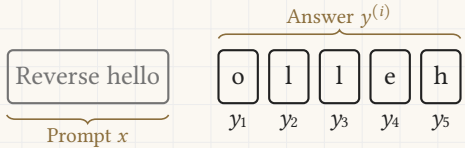
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Every token's return is the answer's reward



Every token's return is the answer's reward



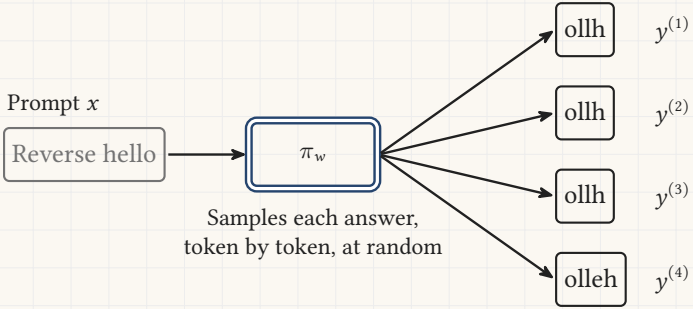
Reward r_t	0	0	0	0	$r^{(i)}$		Every token of $y^{(i)}$: multiplier
Return $G_t = r_t + \cdots + r_5$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$\gamma = 1$	$G_t - \text{baseline} = r^{(i)} - \text{baseline}$

The group: n answers to the same prompt



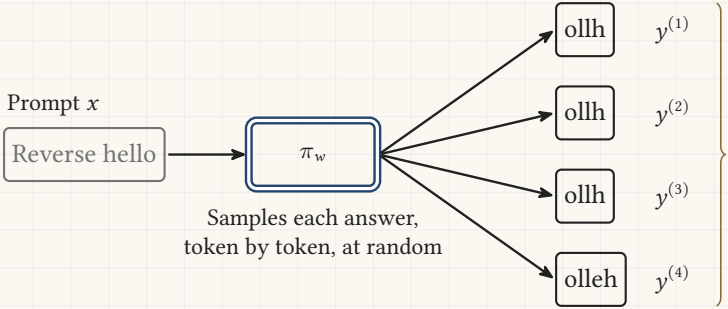
Welleck 2026, CMU CS11-711, lecture 17

The group: n answers to the same prompt



Welleck 2026, CMU CS11-711, lecture 17

The group: n answers to the same prompt



GRPO: Group Relative Policy Optimization

The group of x: its n answers

DeepSeekMath: $n = 64$

Shao et al. 2024, Sec. 4.2

Welleck 2026, CMU CS11-711, lecture 17

The group average is the baseline

Prompt x Reverse hello

Welleck 2026, CMU CS11-711, lecture 17

Answer $y^{(i)}$ | ollh | ollh | ollh | olleh | $n = 4$, sampled from π_w

The group average is the baseline

Prompt x Reverse hello

Welleck 2026, CMU CS11-711, lecture 17

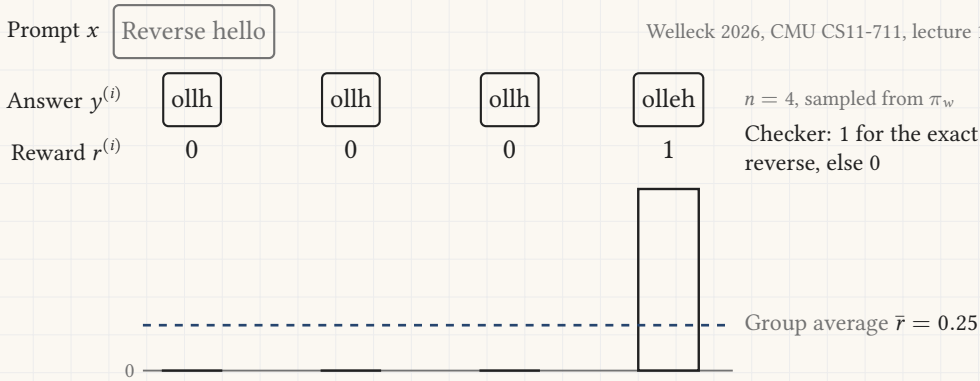
Answer $y^{(i)}$	ollh	ollh	ollh	olleh
Reward $r^{(i)}$	0	0	0	1

$n = 4$, sampled from π_w

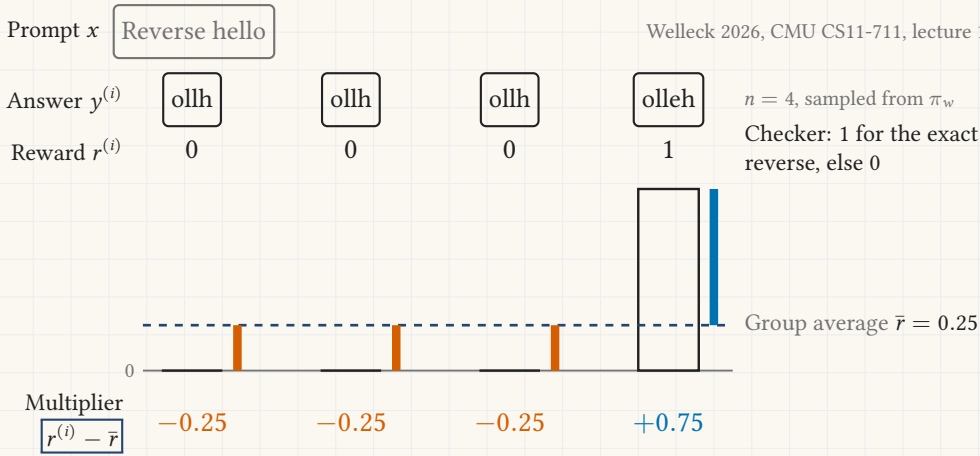
Checker: 1 for the exact reverse, else 0



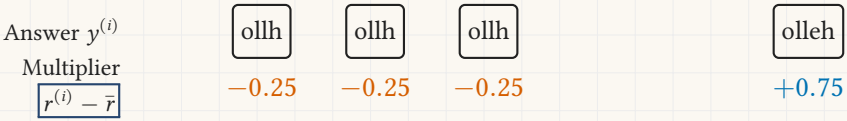
The group average is the baseline



The group average is the baseline

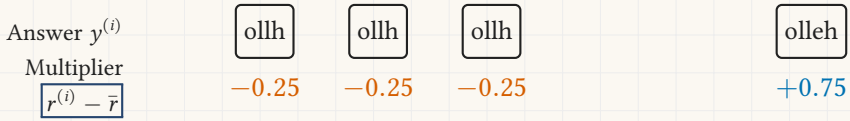


One gradient step on this group



Welleck 2026, CMU CS11-711, lecture 17

One gradient step on this group



$$\tilde{\mathcal{J}} = \frac{1}{4}\Big[\underbrace{3 \cdot (-0.25)\, \log \pi_w(\text{ollh} \mid x)}_{\text{Three answers ollh}} \Big]$$

$\tilde{\mathcal{J}}$: the morning's average of multiplier $\times$ log-probability

Welleck 2026, CMU CS11-711, lecture 17

One gradient step on this group

Answer $y^{(i)}$	ollh	ollh	ollh	olleh
Multiplier $r^{(i)} - \bar{r}$	-0.25	-0.25	-0.25	$+0.75$

$$\tilde{\mathcal{J}} = \frac{1}{4}\Big[\underbrace{3 \cdot (-0.25)\, \log \pi_w(\text{ollh} \mid x)}_{\text{Three answers ollh}} + \underbrace{0.75\, \log \pi_w(\text{olleh} \mid x)}_{\text{One answer olleh}}\Big]$$

$\tilde{\mathcal{J}}$: the morning's average of multiplier $\times$ log-probability

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One gradient step on this group

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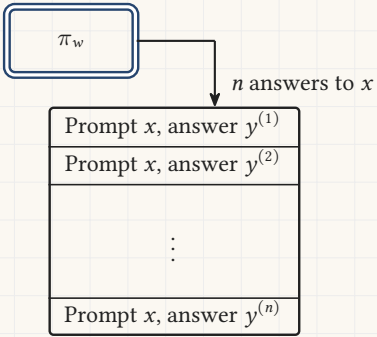
$$\tilde{\mathcal{J}} = \tfrac{1}{4}\Big[\underbrace{3 \cdot (-0.25)\, \log \pi_w(\text{ollh} \mid x)}_{\text{Three answers ollh}} + \underbrace{0.75\, \log \pi_w(\text{olleh} \mid x)}_{\text{One answer olleh}}\Big]$$

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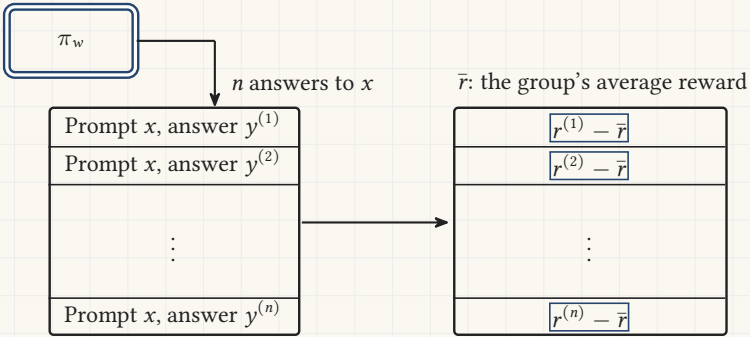
Step along $\nabla_w \tilde{\mathcal{J}}$: ↓ Less likely ↑ More likely

Welleck 2026, CMU CS11-711, lecture 17

The same loop for GRPO

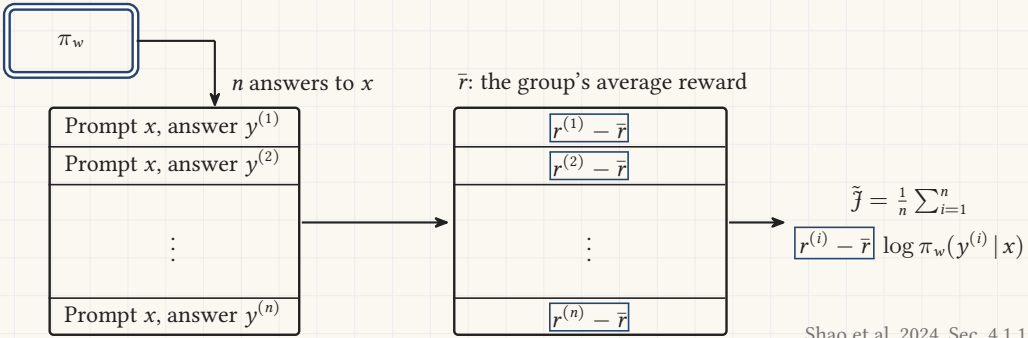


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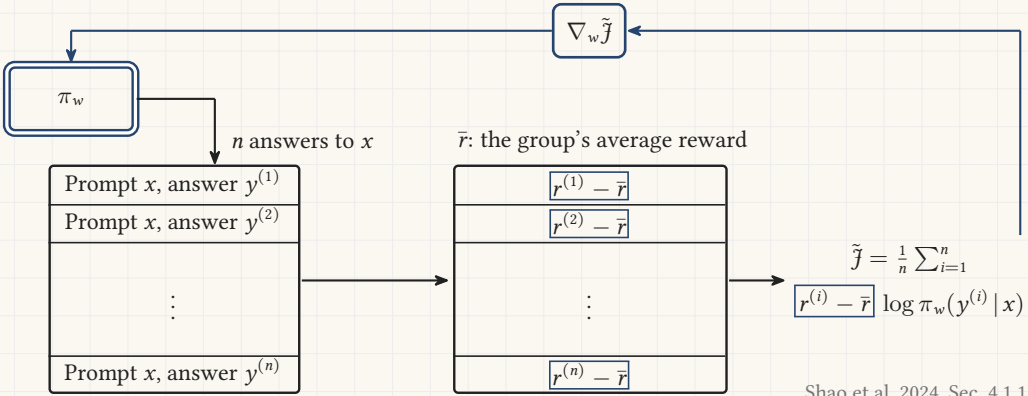
Shao et al. 2024, Sec. 4.1.1

The same loop for GRPO



Shao et al. 2024, Sec. 4.1.1

The same loop for GRPO



Shao et al. 2024, Sec. 4.1.1

Leaving the answer out only rescales the multiplier

Morning's rule: no baseline may see the answer it scores

$\bar{r}$ includes it; RLOO uses $\bar{r}_{-i}$, the others' average



Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

Leaving the answer out only rescales the multiplier

Morning's rule: no baseline may see the answer it scores

$\bar{r}$ includes it; RLOO uses $\bar{r}_{-i}$, the others' average

Answer $y^{(i)}$	ollh	ollh	ollh	olleh
RLOO $r^{(i)} - \bar{r}_{-i}$	-0.33	-0.33	-0.33	$+1$
GRPO $r^{(i)} - \bar{r}$	-0.25	-0.25	-0.25	$+0.75$

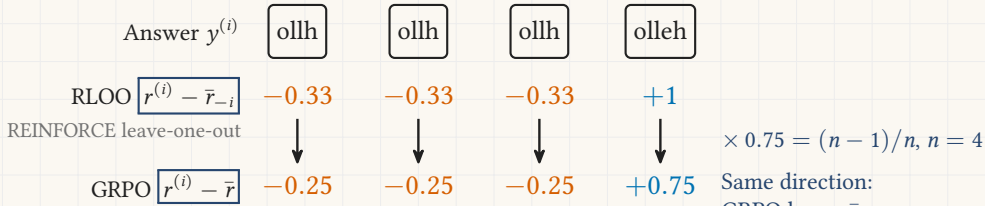
REINFORCE leave-one-out

Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

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Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

The GRPO objective

Shao et al. 2024, eq. (3)

$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\substack{\text{Over answers, then tokens:} \\ \text{divided by the answer's length}}} \Big[\qquad \Big]$$

The GRPO objective

Shao et al. 2024, eq. (3)

$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\substack{\text{Over answers, then tokens:} \\ \text{divided by the answer's length}}} \Big[\underbrace{\min\Big(\rho_t \overbrace{\hat{A}^{(i)}}^{\substack{\text{Group advantage:} \\ \text{divided by the spread}}}, \ \text{clip}(\rho_t, \, 1-\epsilon, \, 1+\epsilon) \, \hat{A}^{(i)}\Big)}_{\text{PPO clipping}} \Big]$$

The GRPO objective

Shao et al. 2024, eq. (3)

$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\substack{\text{Over answers, then tokens:} \\ \text{divided by the answer's length}}} \Big[\underbrace{\min\Big(\rho_t \overbrace{\hat{A}^{(i)}}^{\substack{\text{Group advantage:} \\ \text{divided by the spread}}}, \ \text{clip}(\rho_t, \, 1-\epsilon, \, 1+\epsilon) \, \hat{A}^{(i)}\Big)}_{\text{PPO clipping}} - \underbrace{\beta \, k_t}_{\substack{\text{KL estimate,} \\ \text{weight } \beta}} \Big]$$

The GRPO objective

Shao et al. 2024, eq. (3)

One update per batch: $\rho_t = 1 \Rightarrow$ policy gradient

Group advantage: divided by the spread

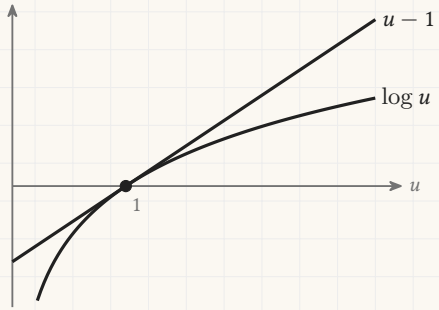
$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\text{Over answers, then tokens: divided by the answer's length}} \Big[\underbrace{\min\big(\rho_t \, \boxed{\hat{A}^{(i)}}, \ \text{clip}(\rho_t, \, 1-\epsilon, \, 1+\epsilon) \, \boxed{\hat{A}^{(i)}}\big)}_{\text{PPO clipping}} - \underbrace{\beta \, k_t}_{\text{KL estimate, weight } \beta} \Big]$$

The KL estimate in the loss

$$k_t = (u - 1) - \log u \quad u = \pi_{\text{ref}}/\pi_{\text{w}}, \text{ per token}$$

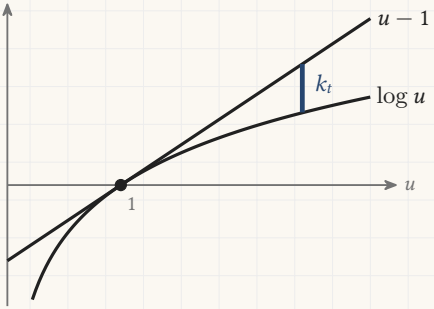
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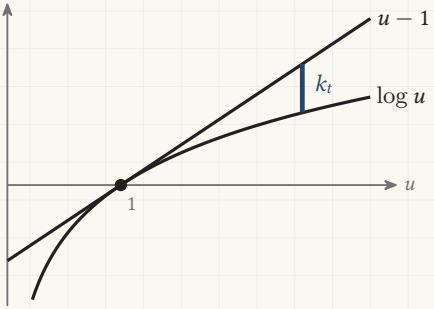


Never negative; unbiased for tokens from π_w

Schulman 2020

The KL estimate in the loss

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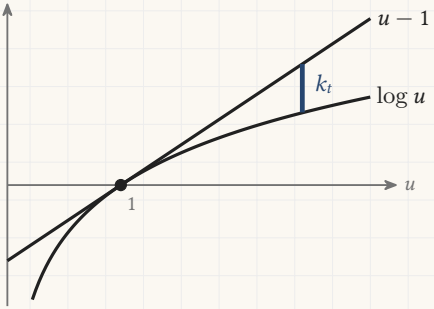
Never negative; unbiased for tokens from π_w

Schulman 2020

Weight β	
DeepSeekMath	0.04
DeepSeek-R1-Zero	0.001 New π_{ref} every 400 steps
Two later variants	0

The KL estimate in the loss

$$k_t = (u - 1) - \log u \quad u = \pi_{\text{ref}}/\pi_w, \text{ per token}$$



Never negative; unbiased for tokens from π_w

Schulman 2020

Weight β

DeepSeekMath	0.04
DeepSeek-R1-Zero	0.001
	New π_{ref} every 400 steps
Two later variants	0

In the reward: may penalise length

Guo et al. 2025

Two later variants: Dr. GRPO and DAPO

Dr. GRPO

GRPO Done Right

Problem	GRPO's two divisions bias it
Change	Drop both
Gain	Same accuracy, shorter wrong answers

Liu et al. 2025

Two later variants: Dr. GRPO and DAPO

	Dr. GRPO GRPO Done Right	DAPO
Problem	GRPO's two divisions bias it	Agreeing groups teach nothing
Change	Drop both	Drop them, refill the batch
Gain	Same accuracy, shorter wrong answers	AIME 2024: 42 → 50
	Liu et al. 2025	Yu et al. 2025, Table 1

Dividing by the spread removes the reward's units

Rewards		Minus $\bar{r}$ (ollh, olleh)
$r \in \{0, 1\}$	Liu et al. 2025	$-0.25,\ +0.75$
$r \in \{0, 10\}$	Lambert et al. 2025	$-2.50,\ +7.50$
$r \in \{-1, 1\}$	Yu et al. 2025	$-0.50,\ +1.50$

Dividing by the spread removes the reward's units

DeepSeekMath
Shao et al. 2024, Sec. 4.1.2

$$\hat{A}^{(i)} = \frac{r^{(i)} - \bar{r}}{\underbrace{\text{std}}}$$

Spread of the group's rewards

Rewards		Minus $\bar{r}$ (ollh, olleh)
$r \in \{0, 1\}$	Liu et al. 2025	$-0.25,\ +0.75$
$r \in \{0, 10\}$	Lambert et al. 2025	$-2.50,\ +7.50$
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Dividing by the spread removes the reward's units

DeepSeekMath

Shao et al. 2024, Sec. 4.1.2

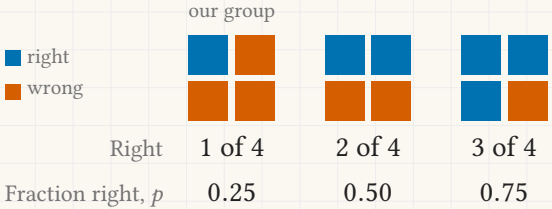
$$\hat{A}^{(i)} = \frac{r^{(i)} - \bar{r}}{\underbrace{\text{std}}_{\text{Spread of the group's rewards}}}$$

Every group gets spread 1:
one step size and one β for any reward

Rewards		Minus $\bar{r}$ (ollh, olleh)	Divided by std: $\hat{A}^{(i)}$	
$r \in \{0, 1\}$	Liu et al. 2025	$-0.25,\ +0.75$	$-0.577,\ +1.732$	
$r \in \{0, 10\}$	Lambert et al. 2025	$-2.50,\ +7.50$	$-0.577,\ +1.732$	
$r \in \{-1, 1\}$	Yu et al. 2025	$-0.50,\ +1.50$	$-0.577,\ +1.732$	std over n

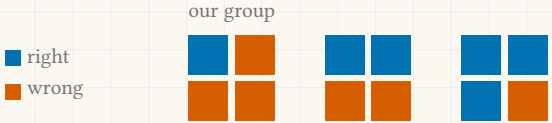
The spread depends only on the fraction right

Three prompts, $n = 4$ answers each, rewards 1 or 0



The spread depends only on the fraction right

Three prompts, $n = 4$ answers each, rewards 1 or 0



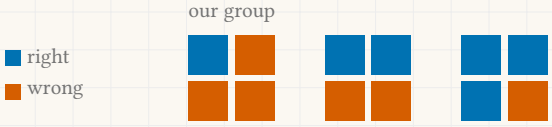
Right	1 of 4	2 of 4	3 of 4
Fraction right, p	0.25	0.50	0.75
Spread, std	0.433	0.500	0.433

$$\text{std} = \sqrt{p\,(1-p)}$$

rewards 1 or 0, std over n

The spread depends only on the fraction right

Three prompts, $n = 4$ answers each, rewards 1 or 0



Right	1 of 4	2 of 4	3 of 4
Fraction right, p	0.25	0.50	0.75
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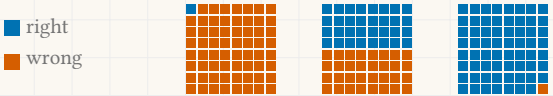
Set by p alone,
and p is how hard the prompt is

few right: hard; most right: easy

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

Shao et al. 2024, Sec. 4.2

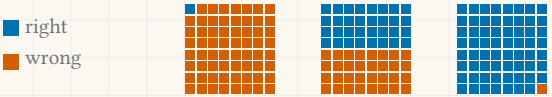


Right	1 of 64	32 of 64	63 of 64
Fraction right, p	0.016	0.500	0.984

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

Shao et al. 2024, Sec. 4.2

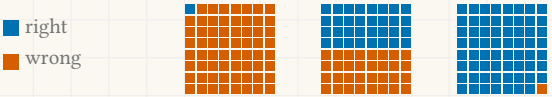


Right	1 of 64	32 of 64	63 of 64
Fraction right, p	0.016	0.500	0.984
Spread, std over n	0.124	0.500	0.124

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

Shao et al. 2024, Sec. 4.2



Right	1 of 64	32 of 64	63 of 64
Fraction right, p	0.016	0.500	0.984
Spread, std over n	0.124	0.500	0.124
Each multiplier $\times$ 1/std	$\times$ 8.06	$\times$ 2.00	$\times$ 8.06

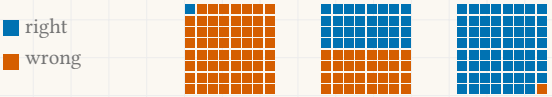
A difficulty bias

Liu et al. 2025, Sec. 3.1

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

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A difficulty bias

Liu et al. 2025, Sec. 3.1

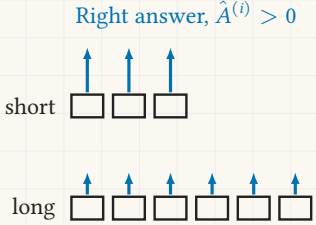
Rewards 1 or 0: no units to remove;
Dr. GRPO drops the division

Dividing by the length favours long wrong answers

GRPO: each token of $y^{(i)}$ gets $\hat{A}^{(i)}/|y^{(i)}|$

Dividing by the length favours long wrong answers

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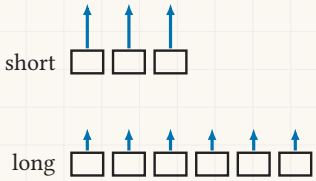
Short ones gain more per token

Dividing by the length favours long wrong answers

GRPO: each token of $y^{(i)}$ gets $\hat{A}^{(i)}/|y^{(i)}|$

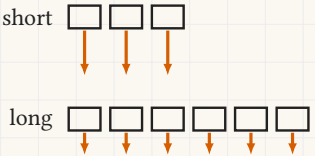
Liu et al. 2025, Sec. 3.1

Right answer, $\hat{A}^{(i)} > 0$



Short ones gain more per token

Wrong answer, $\hat{A}^{(i)} < 0$



Long ones lose less per token:
wrong answers grow longer

Dr. GRPO drops both divisions

Each token of $y^{(i)}$ gets, in GRPO:

$$\left(r^{(i)} - \bar{r}\right) \times \frac{1}{\text{std}} \times \frac{1}{|y^{(i)}|}$$

Dr. GRPO drops both divisions

Each token of $y^{(i)}$ gets, in GRPO:

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Difficulty bias Length bias

Liu et al. 2025, Sec. 3.2

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Difficulty bias Length bias

Liu et al. 2025, Sec. 3.2

In Dr. GRPO:

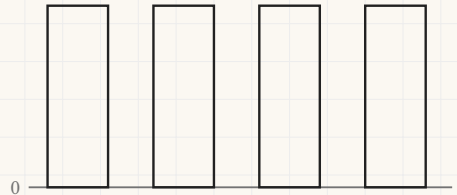
$$\left(r^{(i)} - \bar{r}\right) \times \frac{1}{C}$$

The loop's multiplier, every token alike:
this morning's policy gradient

C: the generation budget, the same for every answer

Groups whose answers all agree teach nothing

Every answer to x correct

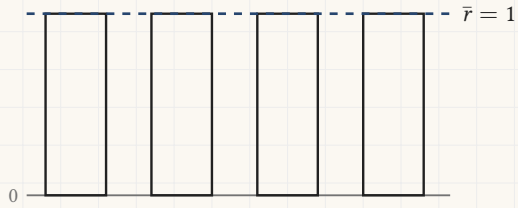


Reward $r^{(i)}$ 1 1 1 1

Groups whose answers all agree teach nothing

Every answer to x correct

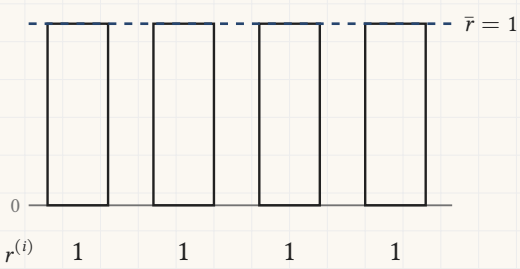
Group average $\bar{r} = 1$: every multiplier $r^{(i)} - \bar{r} = 0$



Groups whose answers all agree teach nothing

Every answer to x correct

Group average $\bar{r} = 1$: every multiplier $r^{(i)} - \bar{r} = 0$

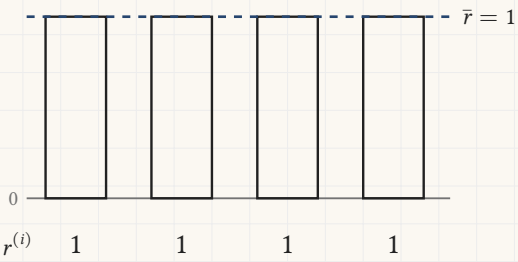


$$\frac{r^{(i)} - \bar{r}}{\text{std}} = \frac{0}{0}$$

Groups whose answers all agree teach nothing

Every answer to x correct

Group average $\bar{r} = 1$: every multiplier $r^{(i)} - \bar{r} = 0$



$$\frac{r^{(i)} - \bar{r}}{\text{std}} = \frac{0}{0}$$

Code adds a small δ:

$$\frac{r^{(i)} - \bar{r}}{\text{std} + \delta} = \frac{0}{\delta} = 0$$

The group adds nothing to the step: its n answers were wasted

Five ways to fill the multiplier

Algorithm	Multiplier	Score	In words
REINFORCE	G_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Rest of the episode
With a baseline	$G_t - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than usual here
Actor-critic	$r_t + \gamma\, V(s_{t+1}) - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than predicted
PPO	Ψ_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Same, step constrained

Five ways to fill the multiplier

Algorithm	Multiplier	Score	In words
REINFORCE	G_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Rest of the episode
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Actor-critic	$r_t + \gamma\, V(s_{t+1}) - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than predicted
PPO	Ψ_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Same, step constrained
GRPO	$(r^{(i)} - \bar{r})/\text{std}$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than its group

std over n or $n-1$

3

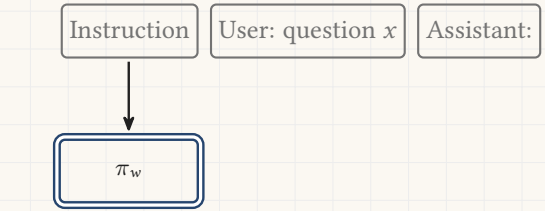
Section 3

DeepSeek-R1

R1-Zero: the template and the reward

Instruction | User: question x | Assistant:

R1-Zero: the template and the reward

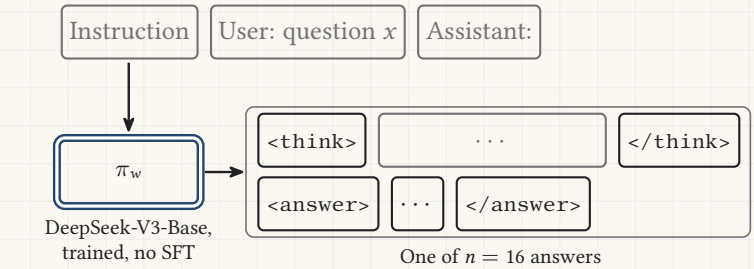


DeepSeek-V3-Base,
trained, no SFT

GRPO, KL weight $\beta = 0.001$
Learning rate 3×10^{-6}

Guo et al. 2025

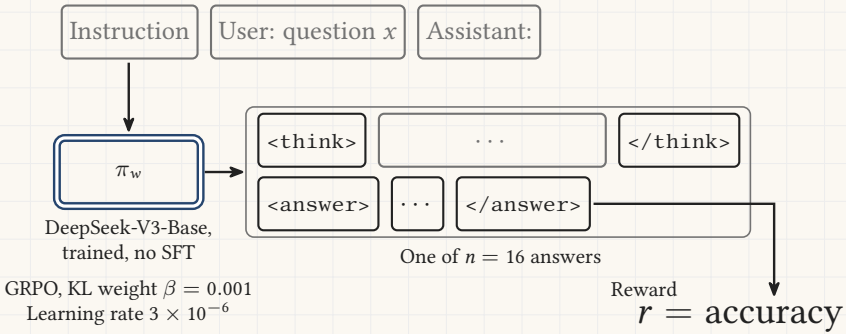
R1-Zero: the template and the reward



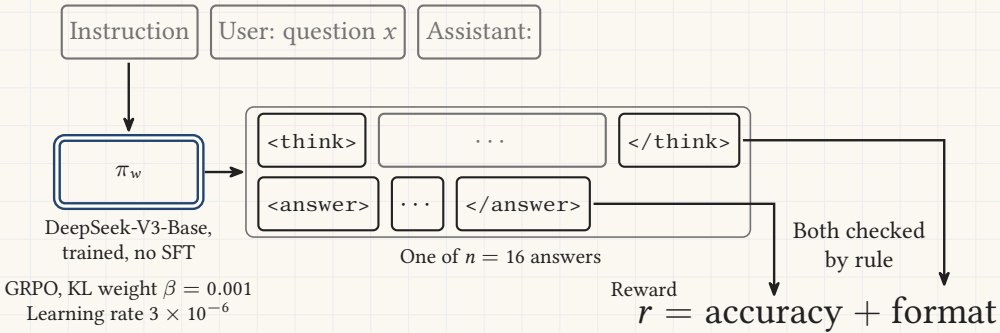
GRPO, KL weight $\beta = 0.001$
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Guo et al. 2025

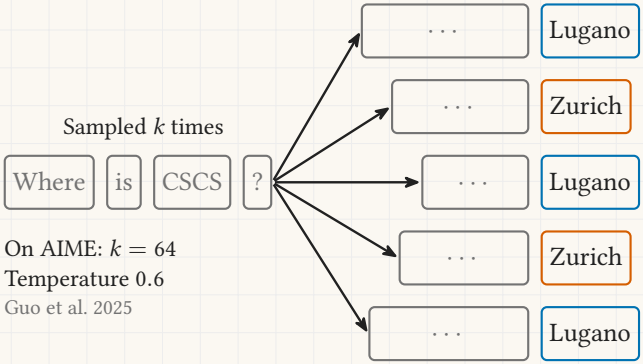
R1-Zero: the template and the reward



R1-Zero: the template and the reward

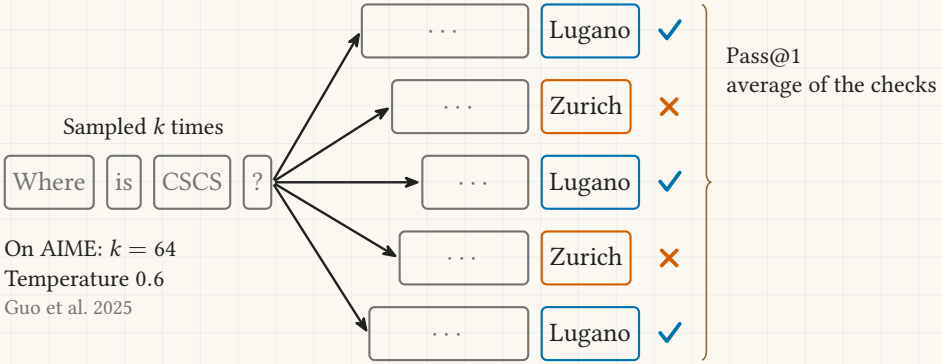


Pass@1, pass@k and majority voting

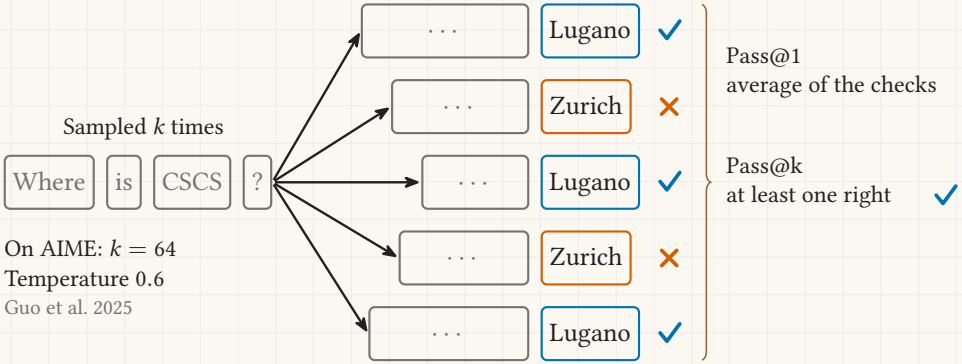


On AIME: $k = 64$
Temperature 0.6
Guo et al. 2025

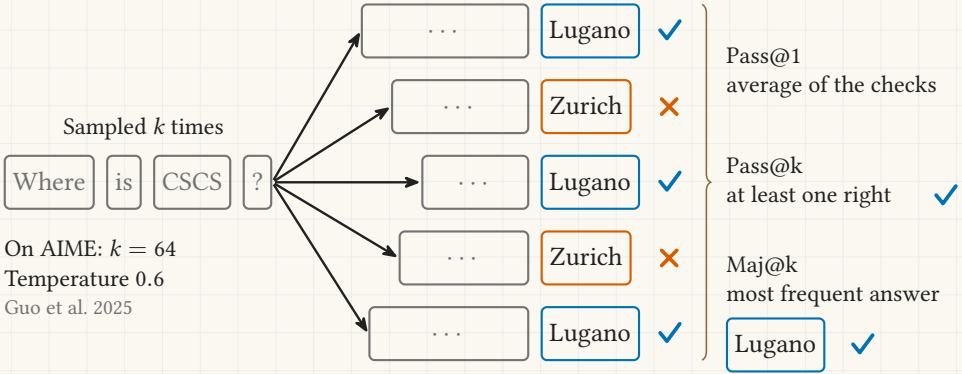
Pass@1, pass@k and majority voting



Pass@1, pass@k and majority voting



Pass@1, pass@k and majority voting

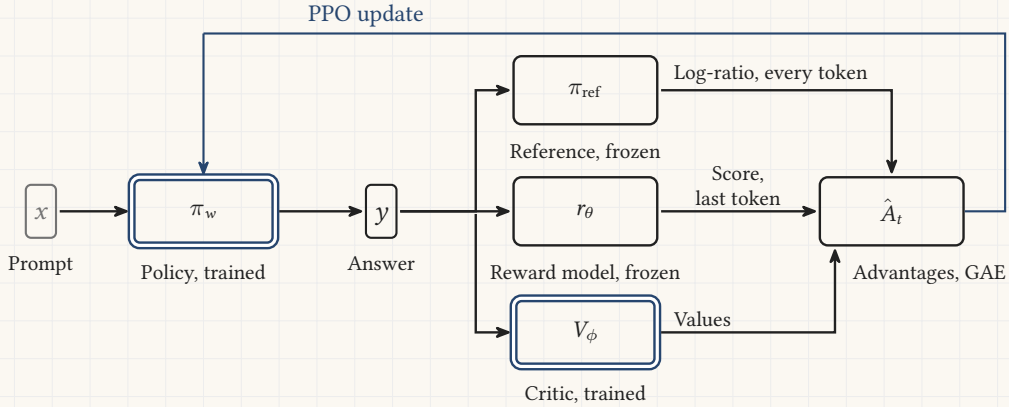


4

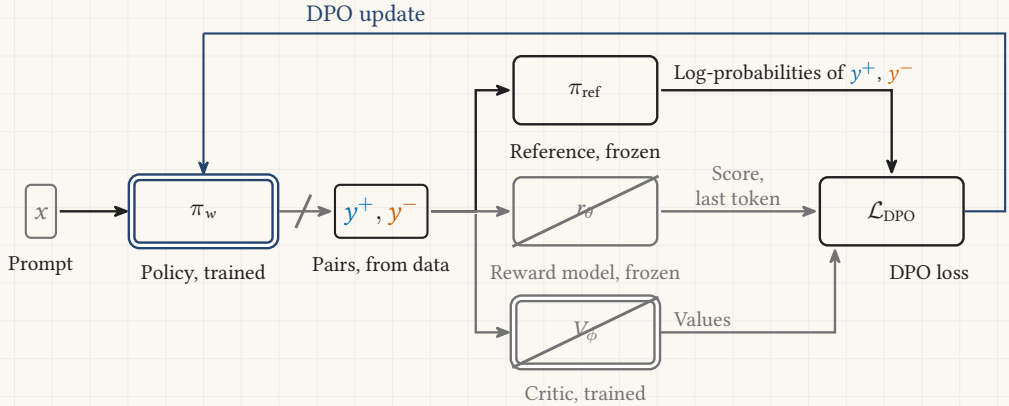
Section 4

Direct preference optimisation

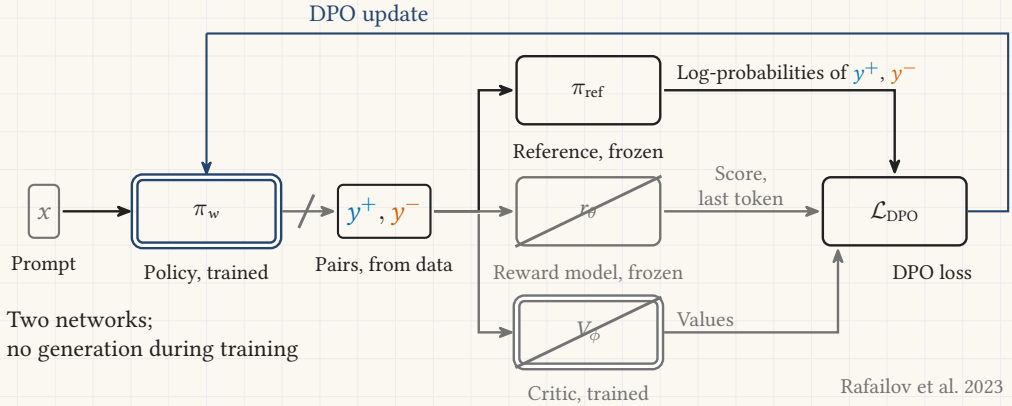
What DPO removes



What DPO removes



What DPO removes

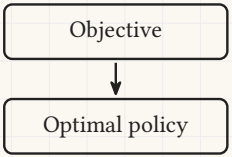


From Part 1's objective to the DPO loss

Objective

$$\mathcal{J}(\pi) = \underset{\text{Answers } y \text{ drawn from any } \pi}{\mathbb{E}}\big[r(x, y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\text{ref}}\big)$$

From Part 1's objective to the DPO loss



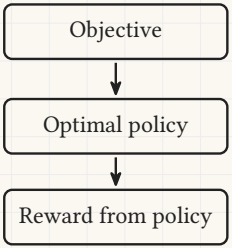
$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

From Part 1's objective to the DPO loss



$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

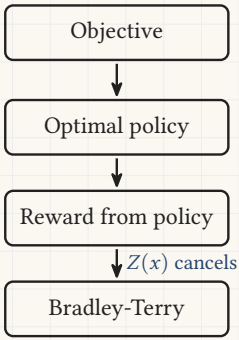
Answers y drawn from any π

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

From Part 1's objective to the DPO loss



$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

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$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

$$P(y^+ \succ y^- \mid x) = \sigma\left(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\mathrm{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\mathrm{ref}}(y^- \mid x)}\right)$$

From Part 1's objective to the DPO loss

Objective

$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

Optimal policy

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

Reward from policy

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

$Z(x)$ cancels

Bradley-Terry

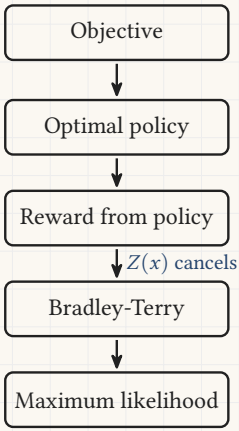
$$P(y^+ \succ y^- \mid x) = \sigma\left(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\mathrm{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\mathrm{ref}}(y^- \mid x)}\right)$$

Maximum likelihood

$$\mathcal{L}_{\mathrm{DPO}}(w) = -\mathbb{E}\big[\log P_w(y^+ \succ y^- \mid x)\big]$$

P_w: π_w for π^*

From Part 1's objective to the DPO loss



$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

Rafailov et al. 2023, eqs. (3) to (7)

Every step: appendix

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

$$P(y^+ \succ y^- \mid x) = \sigma\left(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\mathrm{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\mathrm{ref}}(y^- \mid x)}\right)$$

$$\mathcal{L}_{\mathrm{DPO}}(w) = -\mathbb{E}\big[\log P_w(y^+ \succ y^- \mid x)\big]$$

P_w: π_w for π^*

The DPO loss

Policy, trained

$$\beta \log \frac{\pi_w(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)}$$

The DPO loss

Policy, trained

$$\beta \log \frac{\pi_w(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \beta \log \frac{\pi_w(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)}$$

KL weight

Reference, frozen

The DPO loss

Policy, trained

Sigmoid

$$\log \sigma\left(\beta \log \frac{\pi_w(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \beta \log \frac{\pi_w(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)}\right)$$

KL weight

Reference, frozen

The DPO loss

Rafailov et al. 2023, eq. (7)

$$\mathcal{L}_{\text{DPO}}(w) = -\mathbb{E}\left[\log \sigma\left(\beta \log \frac{\pi_w(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \beta \log \frac{\pi_w(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)}\right)\right]$$

DPO loss

Sigmoid

Policy, trained

Over pairs: prompt x, chosen y^+, rejected y^-

KL weight

Reference, frozen

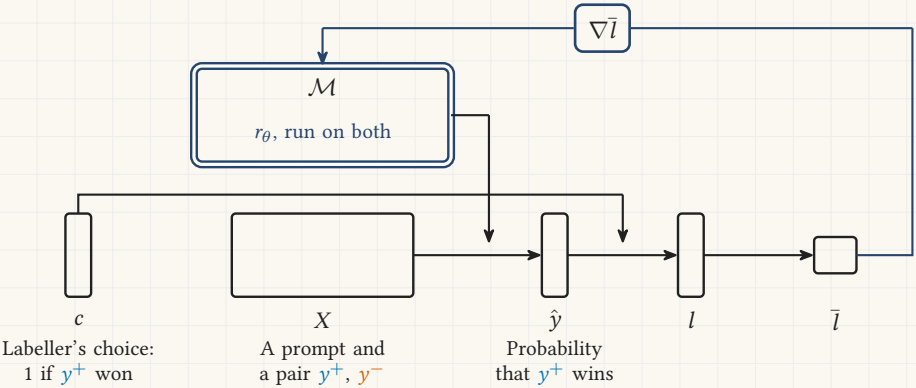
5

Section 5

DPO in practice

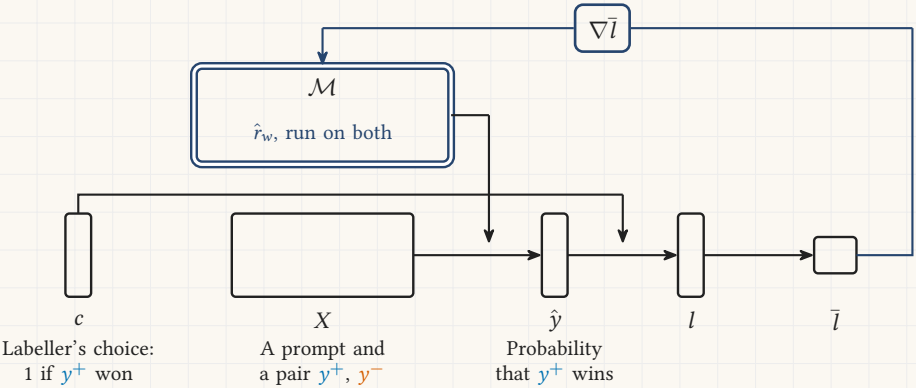
The DPO loss is a reward-model loss

$$\hat{y} = \sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big) \qquad l = -\log \hat{y}$$



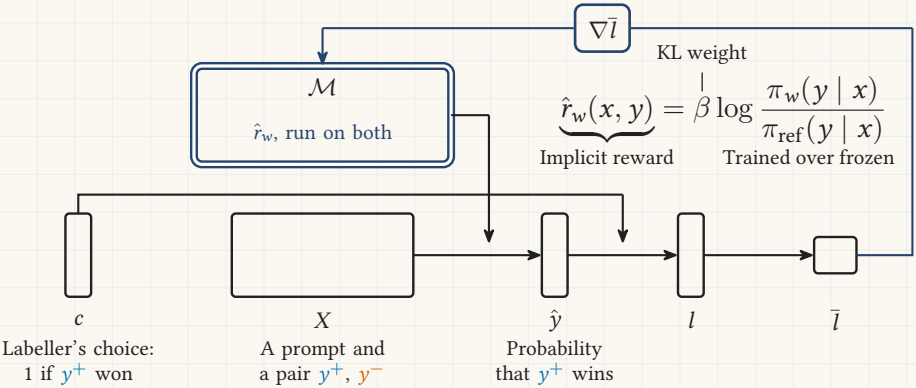
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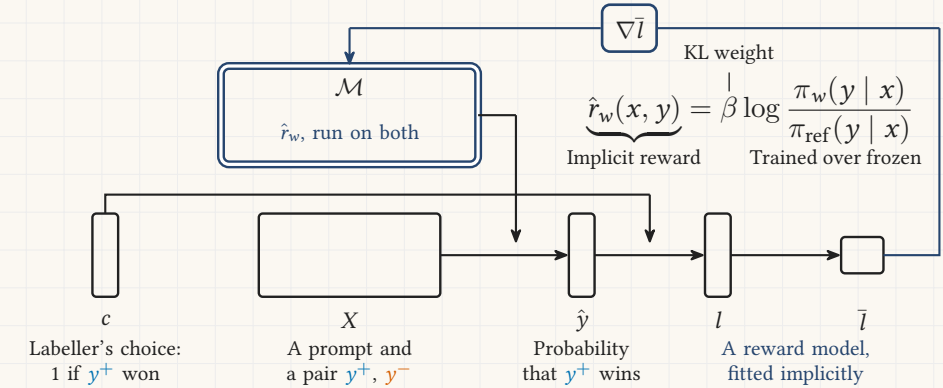
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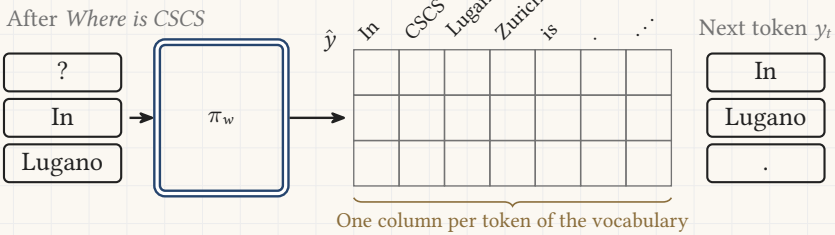


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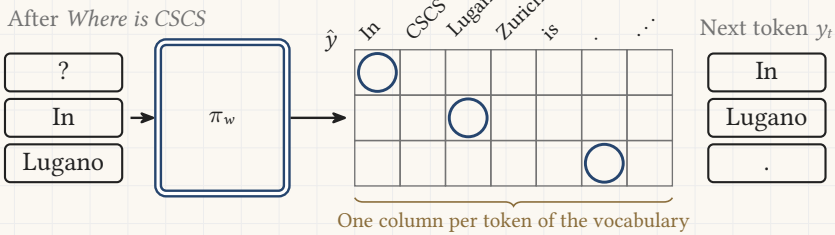
Four log-probabilities



Four log-probabilities

Circled, summed

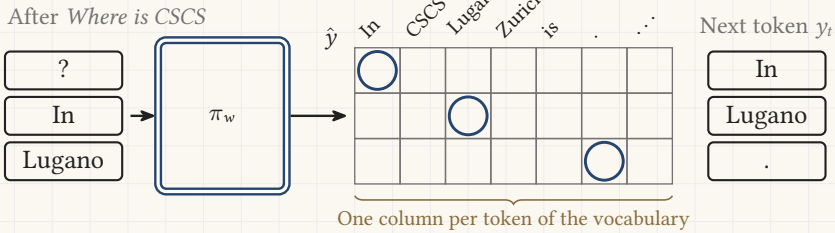
$$\log \pi_w(y^+ \mid x)$$



Four log-probabilities

Circled, summed | Second pass

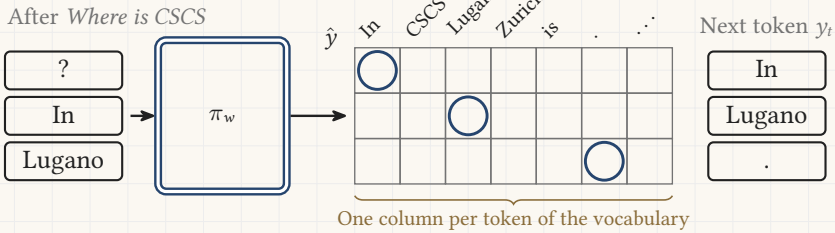
$$\big(\log \pi_w(y^+ \mid x) - \log \pi_w(y^- \mid x)\big)$$



Four log-probabilities

Circled, summed | Second pass | Frozen, cached

$$\big(\log \pi_w(y^+ \mid x) - \log \pi_w(y^- \mid x)\big) - \overbrace{\big(\log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x)\big)}$$



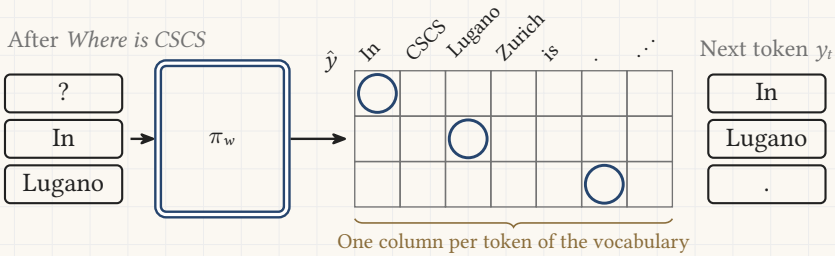
Four log-probabilities

Circled, summed

Second pass

Frozen, cached

$$\underbrace{\beta\Big[\big(\log \pi_w(y^+ \mid x) - \log \pi_w(y^- \mid x)\big) - \overbrace{\big(\log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x)\big)}\Big]}_{\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)}$$



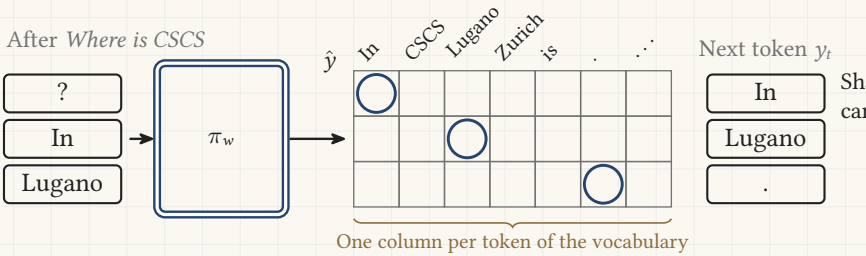
Four log-probabilities

Circled, summed

Second pass

Frozen, cached

$$\underbrace{\beta\Big[\big(\log \pi_w(y^+ \mid x) - \log \pi_w(y^- \mid x)\big) - \overbrace{\big(\log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x)\big)}\Big]}_{\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)}$$



6

Section 6

Comparing the methods

GRPO drops the critic, DPO drops generation

	PPO	GRPO	DPO
Signal	Reward model	Checker or reward model	Pairs

GRPO drops the critic, DPO drops generation

	PPO	GRPO	DPO
Signal	Reward model	Checker or reward model	Pairs
Networks in memory	4	1 to 3	2

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KL penalty	Reward	Loss, or none	Loss
Can be hacked	Reward model	Checker	Pairs

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A

Appendix

Deriving the DPO loss

What the derivation works with

- x One prompt, held fixed
- y An answer: finitely many, bounded length

What the derivation works with

x	One prompt, held fixed
y	An answer: finitely many, bounded length
$\pi(y \mid x)$	Any policy: $\pi(y \mid x) \geq 0$, summing to 1 over y
$\pi_{\text{ref}}(y \mid x)$	The frozen reference: $\pi_{\text{ref}}(y \mid x) > 0$, softmax outputs

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β	The KL weight: $\beta > 0$

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$r(x, y)$	Any reward, a number per answer
β	The KL weight: $\beta > 0$
$\mathbb{E}[f(y)]$	$= \sum_y \pi(y \mid x)\, f(y)$: the average over y drawn from π
$\log$	Natural logarithm; $0 \log 0$ counts 0

Step 1: Part 1's objective as sums

$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x, y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Part 1, one prompt x

Step 1: Part 1's objective as sums

$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\text{ref}}\big)$$

Part 1, one prompt x

$$= \underbrace{\textstyle\sum_y \pi(y \mid x)\, r(x,y)}_{\text{Average reward}} - \beta \underbrace{\textstyle\sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{KL of } \pi \text{ from } \pi_{\text{ref}}}$$

Averages as sums

Step 1: Part 1's objective as sums

Rafailov et al. 2023, eq. (3); App. A.1, eq. (11)

$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\text{ref}}\big)$$

Part 1, one prompt x

$$= \underbrace{\textstyle\sum_y \pi(y \mid x)\, r(x,y)}_{\text{Average reward}} - \beta \underbrace{\textstyle\sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{KL of } \pi \text{ from } \pi_{\text{ref}}}$$

Averages as sums

Each prompt x has its own $\pi(y \mid x)$: solve one prompt at a time.

Step 2: one sum, one logarithm

$$\mathcal{J}(\pi) = \sum_y \pi(y \mid x) \left[r(x, y) - \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \right]$$

Same weights: one sum

Step 2: one sum, one logarithm

$$\mathcal{J}(\pi) = \sum_y \pi(y \mid x) \left[r(x, y) - \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \right]$$

Same weights: one sum

$$= -\beta \sum_y \pi(y \mid x) \left[\log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} - \frac{r(x, y)}{\beta} \right]$$

Take out $-\beta$

Step 2: one sum, one logarithm

$$\mathcal{J}(\pi) = \sum_y \pi(y \mid x) \left[r(x,y) - \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \right]$$ Same weights: one sum

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$$= -\beta \sum_y \pi(y \mid x) \left[\log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} - \log e^{r(x,y)/\beta} \right]$$ $\frac{r}{\beta} = \log e^{r/\beta}$

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Rafailov et al. 2023, App. A.1, eq. (12)

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$$= -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}$$ $\log a - \log b = \log \frac{a}{b}$

Step 3: reweight the reference, then normalise

$$\pi^{*}(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

New policy

Reference

Weight

Step 3: reweight the reference, then normalise

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

New policy

Reference

Weight

$$Z(x) = \sum_y \pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}$$

Normaliser

Step 3: reweight the reference, then normalise

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

New policy · Reference · Weight

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Normaliser

Reference $\pi_{\text{ref}}(y \mid x)$



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New policy · Reference · Weight

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Normaliser

Reference $\pi_{\text{ref}}(y \mid x)$ Weight $e^{r(x,y)/\beta}$



Step 3: reweight the reference, then normalise

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

New policy | Reference | Weight

$$Z(x) = \sum_y \pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}$$

Normaliser

Reference $\pi_{\text{ref}}(y \mid x)$ × Weight $e^{r(x,y)/\beta}$ / $Z(x)$ → New policy $\pi^*(y \mid x)$



$\pi^*(y \mid x) > 0$, as $\pi_{\text{ref}} > 0$

$$\sum_y \pi^*(y \mid x) = \frac{Z(x)}{Z(x)} = 1$$

So π^* is a policy

Step 3: the objective is a KL to π^*

$$\mathcal{J}(\pi) = -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}$$

Step 2

Step 3: the objective is a KL to π^*

$$\mathcal{J}(\pi) = -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}$$

Step 2

$$= -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{Z(x)\, \pi^*(y \mid x)}$$

Definition of π^*

Step 3: the objective is a KL to π^*

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$$= -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{Z(x)\, \pi^*(y \mid x)}$$ Definition of π^*

$$= -\beta \sum_y \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi^*(y \mid x)} + \beta \log Z(x) \sum_y \pi(y \mid x)$$ $\log \frac{a}{bc} = \log \frac{a}{c} - \log b$

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Rafailov et al. 2023, App. A.1, eqs. (12) to (14)

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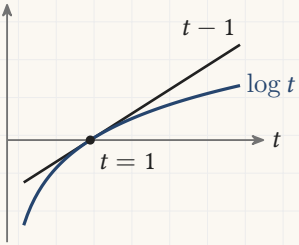
$$= -\beta \underbrace{\mathrm{KL}\big(\pi \,\big\|\, \pi^*\big)}_{\text{KL, as in step 1}} + \underbrace{\beta \log Z(x)}_{\text{Same for every } \pi}$$ $\sum_y \pi(y \mid x) = 1$

Gibbs' inequality: a KL is never negative

For policies p, q of one prompt, with every $q(y) > 0$:
$\mathrm{KL}\big(p \,\|\, q\big) \geq 0$, and $\mathrm{KL}\big(p \,\|\, q\big) = 0$ only if $p = q$.

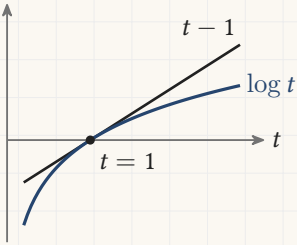
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$t - 1 - \log t$ has slope
$1 - 1/t$: lowest, 0, at $t = 1$.

Gibbs' inequality: a KL is never negative



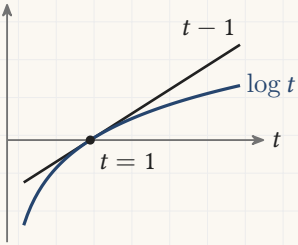
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Terms with $p(y) = 0$ count 0

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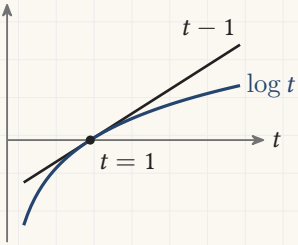
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$$\leq \sum_{p(y)>0} p(y) \left(\frac{q(y)}{p(y)} - 1 \right)$$

$\log t \leq t - 1$

Gibbs' inequality: a KL is never negative



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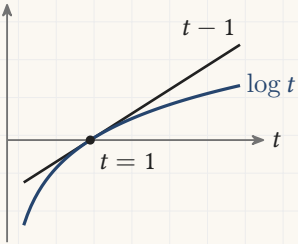
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$\sum_{p(y)>0}p(y)=1$, $\sum_{p(y)>0}q(y)\leq 1$

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$$\leq\sum_{p(y)>0}p(y)\left(\frac{q(y)}{p(y)}-1\right)$$

$\log t\leq t-1$

$$=\sum_{p(y)>0}q(y)-1\;\leq\;0$$

$\sum_{p(y)>0}p(y)=1$, $\sum_{p(y)>0}q(y)\leq 1$

Equal only if $q(y)=p(y)$ wherever $p(y)>0$, and no $p(y)$ is 0, as every $q(y)>0$: so $q=p$.

Step 3: π^* is the only maximiser

$$\mathcal{J}(\pi) = \beta \log Z(x) - \beta \,\mathrm{KL}\big(\pi \,\big\|\, \pi^*\big) \qquad \text{Step 3, as a KL}$$

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$$\mathcal{J}(\pi) = \beta \log Z(x) \text{ exactly when } \pi = \pi^* \qquad \mathrm{KL}\big(\pi^* \,\big\|\, \pi^*\big) = 0\text{; Gibbs: only there}$$

Step 3: π^* is the only maximiser

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$$\mathcal{J}(\pi) = \beta \log Z(x) \text{ exactly when } \pi = \pi^* \qquad \mathrm{KL}\big(\pi^* \,\big\|\, \pi^*\big) = 0;\ \text{Gibbs: only there}$$

The maximiser: $\pi^*(y \mid x) = \dfrac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$

Rafailov et al. 2023, eq. (4)

Small β sharpens π^*; large β keeps it near π_{ref}.

Step 4: reading the reward off π^*

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

Step 3

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$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$ Step 3

$$\log \pi^*(y \mid x) = \log \pi_{\text{ref}}(y \mid x) + \frac{r(x,y)}{\beta} - \log Z(x)$$ Take logs: every factor > 0

Step 4: reading the reward off π^*

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$ Step 3

$$\log \pi^*(y \mid x) = \log \pi_{\text{ref}}(y \mid x) + \frac{r(x,y)}{\beta} - \log Z(x)$$ Take logs: every factor > 0

$$r(x,y) = \underbrace{\beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{Implicit reward}} + \underbrace{\beta \log Z(x)}_{\substack{\text{Same for every} \\ \text{answer to } x}}$$ Solve for r

Step 4: reading the reward off π^*

Rafailov et al. 2023, eq. (5)

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

Step 3

$$\log \pi^*(y \mid x) = \log \pi_{\text{ref}}(y \mid x) + \frac{r(x,y)}{\beta} - \log Z(x)$$

Take logs: every factor > 0

$$r(x,y) = \underbrace{\beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{Implicit reward}} + \underbrace{\beta \log Z(x)}_{\substack{\text{Same for every} \\ \text{answer to } x}}$$

Solve for r

$Z(x)$ sums over all answers: too costly to compute.

Step 5: $Z(x)$ cancels in Bradley-Terry

$$P(y^+ \succ y^- \mid x) = \sigma\big(r(x, y^+) - r(x, y^-)\big)$$

Assumed: Bradley-Terry, reward r; π^* its maximiser

Step 5: $Z(x)$ cancels in Bradley-Terry

$$P(y^+ \succ y^- \mid x) = \sigma\big(r(x, y^+) - r(x, y^-)\big)$$

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$$= \sigma\Big(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} + \beta \log Z(x)$$

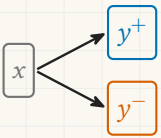
Step 4, for y^+ and for y^-

$$- \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)} - \beta \log Z(x)\Big)$$

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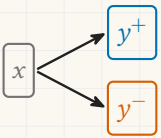
$$- \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)} - \cancel{\beta \log Z(x)}\Big)$$

Same prompt x: $Z(x)$ cancels

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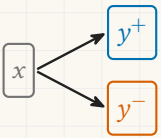
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Rafailov et al. 2023, eqs. (1), (6); App. A.2

$$P(y^+ \succ y^- \mid x) = \sigma\big(r(x, y^+) - r(x, y^-)\big)$$

Assumed: Bradley-Terry, reward r; π^* its maximiser



$$= \sigma\Big(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} + \cancel{\beta \log Z(x)}$$

Step 4, for y^+ and for y^-

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Step 6: the network's implicit reward

$$\hat{r}_w(x, y) = \beta \log \frac{\pi_w(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

Define; $\pi_w > 0$, a softmax

Step 6: the network's implicit reward

$$\hat{r}_w(x, y) = \beta \log \frac{\pi_w(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

Define; $\pi_w > 0$, a softmax

$$\pi_{\text{ref}}(y \mid x)\, e^{\hat{r}_w(x,y)/\beta} = \pi_w(y \mid x)$$

$e^{\log a} = a$

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$$\underset{\text{Optimal for } \hat{r}_w}{\pi^*(y \mid x)} = \frac{\pi_w(y \mid x)}{\sum_{y'} \pi_w(y' \mid x)} = \pi_w(y \mid x)$$

Step 3; the sum is 1

Step 6: the network's implicit reward

$$\hat{r}_w(x, y) = \beta \log \frac{\pi_w(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

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Step 3; the sum is 1

$$\underset{\text{Predicted}}{P_w(y^+ \succ y^- \mid x)} = \sigma\big(\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)\big)$$

Step 5 with this reward

Step 6: the network's implicit reward

Rafailov et al. 2023, Sec. 4

$$\hat{r}_w(x, y) = \beta \log \frac{\pi_w(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

Define; $\pi_w > 0$, a softmax

$$\pi_{\text{ref}}(y \mid x)\, e^{\hat{r}_w(x,y)/\beta} = \pi_w(y \mid x)$$

$e^{\log a} = a$

$$\pi^*(y \mid x) = \frac{\pi_w(y \mid x)}{\sum_{y'} \pi_w(y' \mid x)} = \pi_w(y \mid x)$$

Optimal for $\hat{r}_w$

Step 3; the sum is 1

$$P_w(y^+ \succ y^- \mid x) = \sigma\big(\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)\big)$$

Predicted

Step 5 with this reward

The network is the optimal policy of its own reward.

Step 6: maximum likelihood on the pairs

$$P_w(\text{data}) = \prod_{\text{pairs}} P_w(y^+ \succ y^- \mid x)$$

Independent pairs: multiply

Step 6: maximum likelihood on the pairs

$$P_w(\text{data}) = \prod_{\text{pairs}} P_w(y^+ \succ y^- \mid x)$$ Independent pairs: multiply

$$\log P_w(\text{data}) = \sum_{\text{pairs}} \log P_w(y^+ \succ y^- \mid x)$$ Log of a product: sum of logs

Step 6: maximum likelihood on the pairs

Rafailov et al. 2023, eq. (7)

$$P_w(\text{data}) = \prod_{\text{pairs}} P_w(y^+ \succ y^- \mid x)$$

Independent pairs: multiply

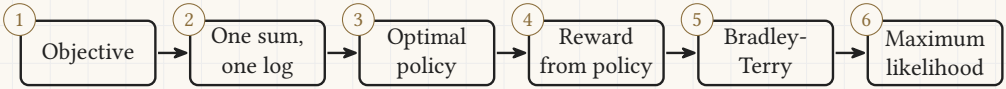
$$\log P_w(\text{data}) = \sum_{\text{pairs}} \log P_w(y^+ \succ y^- \mid x)$$

Log of a product: sum of logs

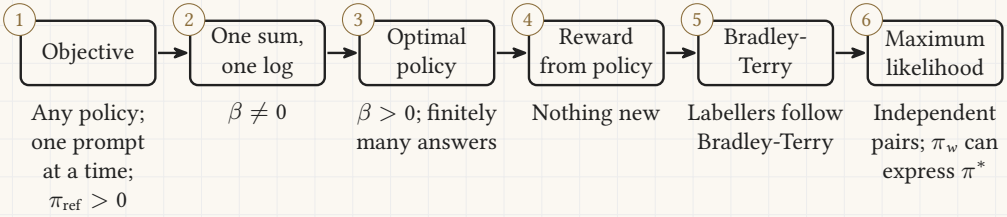
$$\mathcal{L}_{\text{DPO}}(w) = -\frac{1}{N} \underbrace{\sum_{\text{pairs}}}_{\text{The } N \text{ pairs: the } \mathbb{E} \text{ of the DPO loss}} \log P_w(y^+ \succ y^- \mid x)$$

log increases, and $-\frac{1}{N} < 0$: maximise becomes minimise

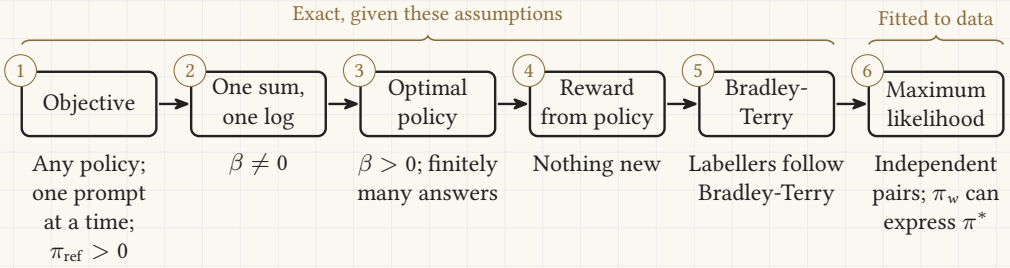
What each step assumes



What each step assumes



What each step assumes



B

Appendix

The group average as a baseline

The group average includes the answer's own reward

This morning's rule

$$\mathbb{E}\big[\, b(x)\, \nabla_w \log \pi_w(y \mid x) \,\big] = b(x)\, \mathbb{E}\big[\nabla_w \log \pi_w(y \mid x)\big] = 0$$

A baseline $b(x)$ may look at the prompt x, never at the answer y

The group average includes the answer's own reward

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$$\mathbb{E}\big[\, b(x)\, \nabla_w \log \pi_w(y \mid x) \,\big] = b(x)\, \mathbb{E}\big[\nabla_w \log \pi_w(y \mid x)\big] = 0$$

A baseline $b(x)$ may look at the prompt x, never at the answer y

Answer $y^{(i)}$	ollh	ollh	ollh	olleh	
Reward $r^{(i)}$	0	0	0	1	
Group average $\bar{r}$	0.25	0.25	0.25	0.25	Sees $y^{(i)}$: breaks the rule

The group average includes the answer's own reward

This morning's rule

$$\mathbb{E}\big[\, b(x)\, \nabla_w \log \pi_w(y \mid x) \,\big] = b(x)\, \mathbb{E}\big[\nabla_w \log \pi_w(y \mid x)\big] = 0$$

A baseline $b(x)$ may look at the prompt x, never at the answer y

Answer $y^{(i)}$	ollh	ollh	ollh	olleh	
Reward $r^{(i)}$	0	0	0	1	
Group average $\bar{r}$	0.25	0.25	0.25	0.25	Sees $y^{(i)}$: breaks the rule
Others' average $\bar{r}_{-i}$	0.33	0.33	0.33	0	Never sees $y^{(i)}$: keeps it

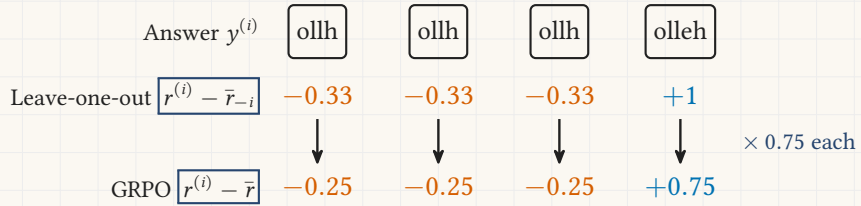
Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

The group average is the others' average, rescaled

Answer $y^{(i)}$	ollh	ollh	ollh	olleh
Leave-one-out $r^{(i)} - \bar{r}_{-i}$	-0.33	-0.33	-0.33	$+1$
GRPO $r^{(i)} - \bar{r}$	-0.25	-0.25	-0.25	$+0.75$

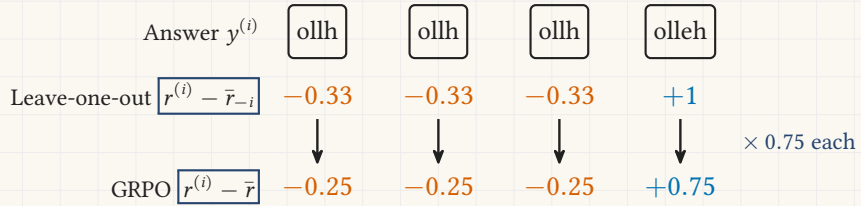
Welleck 2026, CMU CS11-711, lecture 17

The group average is the others' average, rescaled



Welleck 2026, CMU CS11-711, lecture 17

The group average is the others' average, rescaled

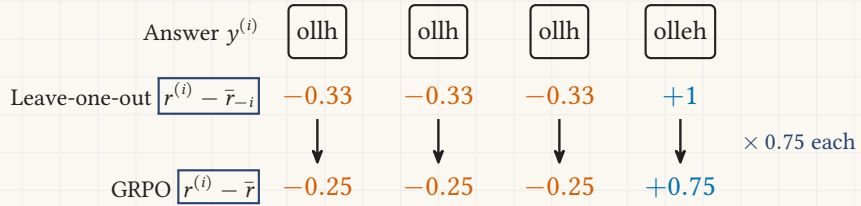


$$r^{(i)} - \bar{r} = \frac{n-1}{n}\left(r^{(i)} - \bar{r}_{-i}\right)$$

since $n\,\bar{r} = r^{(i)} + (n-1)\,\bar{r}_{-i}$

Welleck 2026, CMU CS11-711, lecture 17
Kool et al. 2019; Liu et al. 2025, App. A

The group average is the others' average, rescaled



$$r^{(i)} - \bar{r} = \frac{n-1}{n}\left(r^{(i)} - \bar{r}_{-i}\right)$$

since $n\,\bar{r} = r^{(i)} + (n-1)\,\bar{r}_{-i}$

Leave-one-out: on average the policy gradient
GRPO: times $(n-1)/n$; the step size absorbs it

Welleck 2026, CMU CS11-711, lecture 17
Kool et al. 2019; Liu et al. 2025, App. A